Math 121: Commutative Algebra

Prishita Dharampal

Worksheet 3.1: Gröbner Bases Pt. II

Fix a field $\mathbb{K}$. Unless otherwise specified the ring throughout this worksheet will be $\mathbb{K}[x_1, \ldots, x_n]$. In the previous worksheet we introduced monomial orderings and leading terms. In this worksheet we return to the multivariable division algorithm, and then use it to study monomial ideals, initial ideals, and Gröbner bases. The main goal is to push the story all the way to a major finiteness theorem: every ideal of $\mathbb{K}[x_1, \ldots, x_n]$ is finitely generated.

Recall a *monomial ordering* on $\mathbb{K}[x_1, \ldots, x_n]$ is a total well-ordering $<$ on $\mathbb{Z}^n_{\geq 0}$ such that if $\alpha < \beta$ then $\alpha + \gamma < \beta + \gamma$ for all $\gamma \in \mathbb{Z}^n_{\geq 0}$. Using the bijection between monomials in $\mathbb{K}[x_1, \ldots, x_n]$ and elements of $\mathbb{Z}^n_{\geq 0}$ we think of a monomial ordering as giving us a way to compare monomials by saying $x^\alpha < x^\beta$ if and only if $\alpha < \beta$. In Worksheet 1.3 we saw several important monomial orders: lexicographic (lex) order, graded lexicographic (grlex) order, and graded reverse lexicographic (grevlex) order. Please review that worksheet for their definitions.

Returning to our goal of having a division algorithm in $\mathbb{K}[x_1, \ldots, x_n]$ we are faced with two challenges compared to the familiar setting of $\mathbb{K}[x]$:

(i) In several variables the notion of "leading term" is ambiguous. In one variable, the leading term is simply the term of highest degree. In several variables this no longer suffices: for instance, what should the leading term of $x^2 + xy + y^2$ be? All three terms have total degree 2.

(ii) We want a division algorithm that can answer questions about ideals, i.e., membership, intersections, sums, products, radicals. In $\mathbb{K}[x]$ every ideal is principal, so dividing by a single generator is enough. For $n > 1$, however, $\mathbb{K}[x_1, \ldots, x_n]$ is no longer a PID, so ideals generally require multiple generators. We therefore need a way to divide a polynomial by an entire *list* of polynomials.

As seen on Worksheet 1.3, once a monomial ordering has been fixed, we give a meaningful definition of leading term, leading monomial, and leading coefficient; solving issue (i). Note while we normally write $\mathrm{LT}(f)$, $\mathrm{LM}(f)$ and $\mathrm{LC}(f)$ for these quantities; they do depend on the chosen order $<$, and so would more properly be called $\mathrm{LT}_<(f)$, $\mathrm{LM}_<(f)$ and $\mathrm{LC}_<(f)$ to highlight this fact.

To resolve issue (ii), we generalize long division as follows. In single-variable polynomial long division, we repeatedly ask: "Does the leading term of the divisor divide the current leading term of the dividend?" If yes, we subtract off the appropriate multiple; if not, we are done. The multivariable algorithm works the same way, except that instead of checking a single divisor we scan through an *ordered* list of divisors $G = (g_1, \ldots, g_s)$ and use the first one whose leading term divides the current leading term. More precisely, the algorithm maintains a "working polynomial" p (initialized to f), quotients $q_1, \ldots, q_s$ (initialized to 0), and a remainder r (initialized to 0), then repeats:

- **Try to Divide.** Walk through the list $g_1, g_2, \ldots, g_s$ in order. If you find some g_i whose leading term $\mathrm{LT}(g_i)$ divides $\mathrm{LT}(p)$, subtract the appropriate multiple: update $q_i \leftarrow q_i + \mathrm{LT}(p)/\mathrm{LT}(g_i)$ and $p \leftarrow p - (\mathrm{LT}(p)/\mathrm{LT}(g_i))\, g_i$. Then return to the start of the list and try again with the new p.

- **Move to Remainder.** If *no* g_i in the list has a leading term dividing $\mathrm{LT}(p)$, then the current leading term of p cannot be reduced further. Move it to the remainder: $r \leftarrow r + \mathrm{LT}(p)$ and $p \leftarrow p - \mathrm{LT}(p)$.

We continue until the working polynomial p is equal to 0. Proving this algorithm terminates as expected, which you will do later in this worksheet, amounts essentially to proving the following theorem

Theorem 0.1. *Fix a monomial ordering $<$ on $\mathbb{K}[x_1, \ldots, x_n]$ and let $G = (g_1, \ldots, g_s)$ be an ordered list of nonzero polynomials. If $f \in \mathbb{K}[x_1, \ldots, x_n]$ then there exist $q_1, \ldots, q_s, r \in \mathbb{K}[x_1, \ldots, x_n]$ such that*

$$f = q_1 g_1 + \cdots + q_s g_s + r,$$

where no monomial appearing in r is divisible by any of $LM(g_1), \ldots, LM(g_s)$.

Written in pseudocode the algorithm for multivariable polynomial division is shown below.

Algorithm 1 Division by an ordered list $G = (g_1, \ldots, g_s)$

Require: $f, g_1, \ldots, g_s$
Ensure: $q_1, \ldots, q_s, r$

```
q_1, ..., q_s ← 0; r ← 0; p ← f
while p ≠ 0 do
    divides ← 0
    for i = 1, ..., s do
        if LT(g_i) divides LT(p) then
            q_i ← q_i + LT(p)/LT(g_i)
            p ← p − (LT(p)/LT(g_i))g_i
            divides ← 1
            break
        end if
    end for
    if divides = 0 then
        r ← r + LT(p)
        p ← p − LT(p)
    end if
end while
```

Problem 1. Review of Monomial Orders.
Work in $\mathbb{K}[x, y, z]$ and assume $x > y > z$.

(a) Order the monomials $x^2y,\ xy^2,\ xz^2,\ y^3,\ yz^4$ from largest to smallest with respect to lex, graded lex, and graded reverse lex.

(b) Compute $\mathrm{LM}(f)$, $\mathrm{LC}(f)$, and $\mathrm{LT}(f)$ for $f = 4x^2y - 3xy^2 + y^4 + 2z^5$ with respect to lex and grevlex.

(c) Prove that 1 is the smallest monomial for every monomial ordering.

(d) Deduce that every strictly decreasing sequence of monomials terminates.

(e) Explain why part (d) should make you optimistic that a division algorithm might terminate.

Solution.

(a)
- Lex: x^2y, xy^2, xz^2, y^3, yz^4
- Graded Lex: yz^4, x^2y, xy^2, xz^2, y^3
- Graded RevLex: yz^4, x^2y, xy^2, y^3, xz^2

(b) Lex:
$$\mathrm{LM}(f) = x^2y, \quad \mathrm{LC}(f) = 4, \quad \mathrm{LT}(f) = 4x^2y$$
Grevlex:
$$\mathrm{LM}(f) = z^5, \quad \mathrm{LC}(f) = 2, \quad \mathrm{LT}(f) = 2z^5$$

(c) The monomial 1 corresponds to $(0, \cdots, 0)$. For the sake of contradiction assume there exist some $\alpha = (a_1, \cdots, a_n)$ such that $\alpha \neq 1$ and $\alpha < 1$. Then, by property (ii) of monomial orders we have
$$\alpha + \alpha < 0 + \alpha$$
Applying (ii) again,
$$3\alpha < 2\alpha$$
And so on,
$$\cdots < 4\alpha < 3\alpha < 2\alpha < \cdots$$
But this violates the well-ordering property (iii) of monomial orderings, a contradiction! Hence, 1 must be the smallest monomial in every monomial ordering.

(d) Since we have a minimal element, every strictly decreasing sequence of monomials terminates.

(e) In a division algorithm, at each step we reduce the polynomial f by replacing its leading term with a smaller combination of terms coming from the divisors. This produces a strictly decreasing sequence of leading monomials and by part (d) every such sequence must terminate. Hence, eventually the algorithm must terminate.

Problem 2. Examples of Multivariable Division.
For this question, we will work in $\mathbb{K}[x,y]$ considered with the lex order where $x > y$.

1. Divide $f = x^2y + xy + y - x - 2$ by the ordered list $G = (xy - 1, y - 1)$. Record the successive values of p, the quotients, and the remainder after each update.
2. Divide $f = xy^2 - x$ by the ordered list $G = (xy - 1,\ y^2 - 1)$. Record the successive values of p, the quotients, and the remainder after each update.
3. Repeat (b), but with the ordered list $(y^2 - 1,\ xy - 1)$, i.e. where the order of G is swapped.
4. Show directly that $xy^2 - x \in \langle xy - 1, y^2 - 1\rangle$.
5. What do your computations show about division in several variables? In particular, what is always true when the remainder is 0, and what can go wrong when the remainder is non-zero?

Solution.

1. **Step 0:** $q_1 = 0,\ \ q_2 = 0,\ \ p = x^2y + xy + y - x - 2,\ \ r = 0$

   Step 1: $q_1 = x,\ \ q_2 = 0,\ \ p = xy + y - 2,\ \ r = 0$

   Step 2: $q_1 = xy,\ \ q_2 = 0,\ \ p = y - 1,\ \ r = 0$

   Step 3: $q_1 = xy,\ \ q_2 = y,\ \ p = 0,\ \ r = 0$

2. **Step 0:** $q_1 = 0,\ \ q_2 = 0,\ \ p = xy^2 - x,\ \ r = 0$

   Step 1: $q_1 = y,\ \ q_2 = 0,\ \ p = -x + y,\ \ r = 0$

   Step 2: $q_1 = y,\ \ q_2 = 0,\ \ p = y,\ \ r = -x$

   Step 3: $q_1 = y,\ \ q_2 = 0,\ \ p = 0,\ \ r = -x + y$

3. **Step 0:** $q_1 = 0,\ \ q_2 = 0,\ \ p = xy^2 - x,\ \ r = 0$

   Step 1: $q_1 = x,\ \ q_2 = 0,\ \ p = 0,\ \ r = 0$

4. From part 3 we know that $xy^2 - x \in \langle y^2 - 1\rangle \implies xy^2 - x \in \langle xy - 1, y^2 - 1\rangle$.

5. If the remainder is zero the polynomial is in the ideal generated by the divisors. The converse is not true; that is if the remainder is non-zero, we can't conclude that the polynomial is not in the ideal.

Problem 3. Termination of Division Algorithm
Let $G = (g_1, \ldots, g_s)$ be an ordered list of non-zero polynomials.

1. Show that throughout the algorithm one always has $f = q_1 g_1 + \cdots + q_s g_s + p + r$.

2. Show that after each pass through the `while` loop, the leading monomial of p strictly decreases with respect to the given monomial order.

3. Use Question 1(d) to prove that the algorithm terminates.

4. Prove that on termination the output satisfies $f = q_1 g_1 + \cdots + q_s g_s + r$, and no monomial appearing in r is divisible by any of $\mathrm{LM}(g_1), \ldots, \mathrm{LM}(g_s)$.

5. Explain why a zero remainder certifies ideal membership: if dividing f by G produces remainder 0, then $f \in \langle g_1, \ldots, g_s \rangle$.

Solution.

1. We will show this by induction on the number of steps.

   Base Case: It holds initially since $p = f$ and $q_i = r = 0$.

   Inductive Step: Assume that the the invariant $f = q_1 g_1 + \cdots + q_s g_s + p + r$ is maintained for the first $n-1$ steps, then at the n-th step, we either,

   - Replace $q_i \leftarrow q_i + \frac{\mathrm{LT}(p)}{\mathrm{LT}(g_i)}$ and $p \leftarrow p - \frac{\mathrm{LT}(p)}{\mathrm{LT}(g_i)} g_i$, so the sum $q_i g_i + p$ is unchanged.
   - Or replace $r \leftarrow r + \mathrm{LT}(p)$ and $p \leftarrow p - \mathrm{LT}(p)$, so $p + r$ is unchanged.

   In both cases the invariant is preserved, thus we always have $f = q_1 g_1 + \cdots + q_s g_s + p + r$.

2. If a g_i divides the leading term, we multiply it with q_i and subtract the leading term, if no g_i divides the leading term, we subtract it from p (and add it to r), either ways each pass through the `while` loop removes the leading monomial and thus the leading monomial of p strictly decreases with respect to the given monomial order.

3. Since every strictly decreasing sequence of monomials terminates (Question 1(d)), p must also terminate, i.e. $p = 0$ eventually and the algorithm terminates.

4. From part 1 we know that we always have $f = q_1 g_1 + \cdots + q_s g_s + p + r$, since on termination we have $p = 0$ we get,

   $$f = q_1 g_1 + \cdots + q_s g_s + r$$

   Moreover, since the algorithm only adds $\mathrm{LT}(p)$ to r when $\mathrm{LT}(g_i) \nmid \mathrm{LT}(p)$ for all i no monomial in r is divisibile by any of $\mathrm{LM}(g_1), \ldots, \mathrm{LM}(g_s)$.

5. If $r = 0$ we have $f = q_1 g_1 + \cdots + q_s g_s$ which is precisely the condition for ideal membership.